YAHYA MENKARI

AI & MACHINE LEARNING ENGINEER

RAG • Agentic & Workflow Automation • Time-Series Forecasting

CONTACT

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TECHNICAL STACK

Programming
- Python · SQL

AI & Machine Learning
- Machine Learning · Deep Learning
- XGBoost · Time-Series Forecasting
- RAG · Agentic AI

Automation
- n8n · Workflow Automation
- Streamlit · REST APIs

Data & Retrieval
- Neo4j · Qdrant
- Semantic Search · Knowledge Graphs

Cloud & Tools
- Docker · GCP · Git
- Power BI

LANGUAGES

- **Arabic — Native**
- **French — Bilingual**
- **English — B2+**

CREDENTIALS

Databricks Fundamentals Accreditation — 2026

PROFILE

AI & Machine Learning Engineer with hands-on experience building end-to-end ML, RAG, and automation systems. Developed a 24-hour forecasting pipeline using 43,000+ real-world observations and AI applications combining graph/vector retrieval, multi-agent workflows, and cloud deployment.

EXPERIENCE

Machine Learning Research Intern | Technical University of Sofia, Bulgaria [View in Github] **| Feb 2026 – Jun 2026**

- Developed an end-to-end forecasting pipeline using **43,000+ hourly observations** collected **over five years**.
- Engineered **43 predictive features** using cyclical encodings, lag variables, and rolling statistics.
- Benchmarked **XGBoost, GRU, CNN-LSTM, CNN-GRU, and TCN** for 24-hour forecasting; optimized **XGBoost achieved MAE 2.34, RMSE 3.05, and R^2 0.91** on unseen data.

AI Research Intern | Archipel Cognitive, Morocco | Jul 2024 – Sep 2024

- Researched and designed **RAG architectures for social-media content retrieval**, covering ingestion, embeddings, vector databases, semantic retrieval, and LLM integration.
- Developed a **multi-stage X/Twitter content extraction workflow** and investigated **CDC-based streaming retrieval** for incremental vector updates and **Agentic RAG**

SELECTED PROJECTS

AI-Powered Job Discovery & Outreach Automation Platform | *n8n* | *Python* | *Streamlit* | *Docker* | *GCP*

- Designed and deployed a **multi-stage automation platform** for job discovery, qualification, contact research, outreach preparation, and application tracking.
- Built modular **n8n workflows** with conditional routing, deduplication, persistent state management, and a **Streamlit dashboard** deployed in **Docker on GCP.**

Graph-Based AI Symptom Checker (*View GitHub*) |*Neo4j, Qdrant, RAG, PubMed*

- Built a knowledge-graph and semantic retrieval system combining Neo4j, Qdrant, PubMed, and RAG.

Medical AI Assistant with Specialist Referral System(*View GitHub*) |*CrewAI, Gemini, Streamlit*

- Developed a multi-agent healthcare assistant automating specialist referral, report generation, and information retrieval workflows.

EDUCATION

International University of Rabat (UIR), Morocco

- M.Eng. in Computer Science – Big Data & Artificial Intelligence (2021 – 2026) – Mention Très Bien

Technical University of Sofia, Bulgaria

- Exchange Program – IT for Business Management (2025 – 2026)